Tutorial 7 and 8

Q1

Three security goals are __c__ .

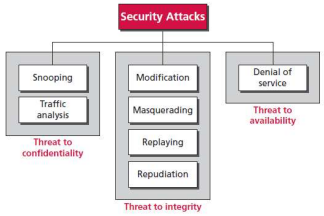
a) confidentiality, cryptography, and nonrepudiation
b) confidentiality, encryption, and decryption
c) confidentiality, integrity, and availability
d) confidentiality, denial of service, and masquerading

- *Non-repudiation is **the assurance that someone cannot deny the validity of something**. Non-repudiation is a legal concept that is widely used in information security and refers to a service, which provides proof of the origin of data and the integrity of the data.*
- *Masquerading to assume the appearance of something one is not*

Q2

a Which of the following attacks is threatening integrity?

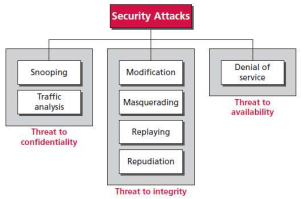
a) Masquerading (pretend to be someone one is not)
b) Traffic Analysis
c) Denial of service
d) Encoding



Q3

c Which of the following attacks is threatening availability?

a) Replaying
b) Modification
c) Denial of service
d) Decoding



Q4

_____b_____ means concealing the contents of a message by enciphering.

a) Steganography 隐写术
b) Cryptography 密码学
c) Compressing
d) Authentication

Q5

_____b_____means concealing the message by covering it with something else.

a) Cryptography
b) Steganography
c) Compressing
d) Authentication

Q6

In _____a_____ cryptography, the same key is used by the sender and the receiver.

a) symmetric-key 对称钥匙
b) asymmetric-key
c) public-key
d) open-key

Q7

In _____a_____ cryptography, the same key is used in both directions.

a) symmetric-key
b) asymmetric-key
c) public-key
d) open-key

Q8

_____a_____ cryptography is often used for long messages.

a) Symmetric-key
b) Asymmetric-key
c) Public-key
d) Open-key

Q9

_____b_____ cryptography is often used for short messages.

a) Symmetric-key
b) Asymmetric-key
c) Secret-key
d) Open-key

Q10

_____d_____means that the sender and the receiver expect confidentiality. 保密性

a) Nonrepudiation
b) Integrity
c) Authentication
d) encryption and decryption 加密和解密

Q11

_____b_____ means that the data must arrive at the receiver exactly as they were sent. 保持信息的不变性

a) Nonrepudiation
b) Message integrity 即信息的完整性
c) Authentication
d) Secrecy

Q12

a _____ can provide authentication, integrity, and nonrepudiation for a message.

a) Encryption/decryption
b) Digital signature
c) Compression
d) Key-exchange

Q13

C In _____, the identity of a party is verified once for the entire duration of system access.

a) entity authentication
b) message integrity
c) message authentication
d) message encryption

实体认证：一次认证多条消息
消息认证：一次认证一条特定消息

Q14

a In _____ cryptography, everyone has access to everyone's public key.

a) symmetric-key
b) asymmetric-key
c) secret-key
d) private-key

Q15

a In the asymmetric-key method used for confidentiality, which key(s) is (are) publicly known?

a) encryption key only
b) decryption key only
c) both encryption and decryption keys
d) neither encryption key nor decryption key

Q16

a

- The RSA algorithm for confidentiality uses _____ cryptography.
- asymmetric-key
- symmetric-key
- substitution
- transposition

Knowledge and Application

Q1

Which of the following words mean ‘secret writing’? Which one means ‘covered writing’?

a. cryptography
b. Steganography

Cryptography means "secret writing." Steganography means "covered writing."

Q2

Alice has found a way to write secretly to Bob. Each time, she takes a new text, such as an article from the newspaper, but inserts one or two spaces between the words. A single space means a binary digit 0; a double space means a binary digit 1. Bob extracts the binary digits and interprets them using ASCII code. Is this an example of cryptography or steganography? Explain.

This is an example of steganography because the message is hidden inside the essay.

Q3

Alice and Bob exchange confidential messages. They share a very large number as the encryption and decryption key in both directions. Is this an example of symmetric-key or asymmetric-key cryptography? Explain.

This is an example of symmetric-key cryptography because they share the same secret key and use the same key for communication in both directions, the meaning of symmetric.

Q4

Distinguish between a substitution cipher and a transposition cipher.

In a substitution cipher a set of characters (or bits) in the plaintext is substituted by another set of characters (or bits) in the ciphertext; a transposition cipher changes the positions of the characters (or bits) in the plaintext to create a ciphertext.

Q5

In a cipher, all the As in the plaintext have been changed to Ds in the ciphertext and all the Ds in the plaintext have been changed to Hs in the ciphertext. Is this a monoalphabetic or polyalphabetic substitution cipher? Explain.

单字母替换密码：变化基于字母本身

This is a monoalphabetic substitution cipher because changing depends on what the character is, not on the position of the character.

Q6

monoalphabetic更好破译

Which cipher can be broken more easily, monoalphabetic or polyalphabetic?

Definitely, a monoalphabetic cipher can be broken more easily. For example, if we find that the decryption of x is a, then all x's are decrypted as a's. This is not true with polyalphabetic ciphers.

Q7

Assume Alice and Bob use an additive cipher in modulo 26 arithmetic. If Eve, the intruder, wants to break the code by trying all possible keys (brute-force attack), how many keys should she try on average?

The key in this case needs to be between 0 and 25 (although there is no cipher if the key is 0). On average Eve needs to test 26/2 = 13 keys to break the code.

Q8

Why do you think asymmetric-key cryptography is used only with small messages?

Because the keys for asymmetric-key is large

To be safe, the keys used in asymmetric-key cryptography should be very large. This makes the calculation very long if the message is also long. Compare the short key of DES (56 bits) with the long key of RSA (500 to 1000 bits).

Q9

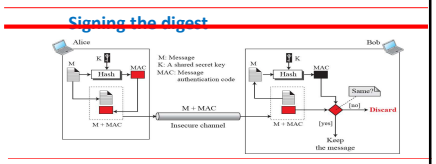
In an asymmetric public key cipher, which key is used for encryption? Which key is used for decryption?

- **a.** public key
- **b.** private key

In an asymmetric public key, the sender uses the public key of the receiver for encryption. The receiver uses her private key for decryption.

Q10

What is the role of the secret key added to the hash function in Figure (on slide 11, week 8, part2) in the text (MAC)? Explain.



The secret key eliminates the need for a secure channel to send the digest. In other words, the secret key prevents Eve from intercepting the digest and changing it to her benefit.

Q11

Distinguish between message authentication and entity authentication.

Message authentication and entity authentication both authenticate the sender for the receiver. As their names indicate, however, message authentication authenticates the sender for a particular message, while entity authentication authenticates the sender for the entire session in which several messages can be sent. If the sender is sending only a single message, message authentication has the same effect as entity authentication; if a sender is sending several messages, one entity authentication authenticates the sender for all messages.

Q12

Which of the following services are not provided by digital signature?

a. message authentication
b. confidentiality
c. Nonrepudiation

Digital signature cannot provide confidentiality.

Q1

Define the type of attack in each of the following cases:

a. A student breaks into a professor's office to obtain a copy of the next test.
b. A student gives a check for $10 to buy a used book. Later the student finds out that the check was cashed for $100.
c. A student sends hundreds of emails per day to the school using a phony return email address.

a. This is **snooping** (an attack to confidentiality). Although the contents of the test are not confidential on the day of the test, they are confidential before the test day.
b. This is **modification** (an attack to integrity). The value of the check is changed (from $10 to $100).
c. This is **denial of service** (an attack to availability). Sending so many emails may crash the server and the service may be interrupted.

Q2

Use the additive cipher with k =10 to encrypt the plaintext 'book'. Then decrypt the message to get the original plaintext.

The following shows the character-by-character encryption of the plaintext.

The ciphertext is "**LYYU**".

Plaintext	Encryption	Ciphertext
b → 01	(01 + 10) mod 26	11 → L
o → 14	(14 + 10) mod 26	24 → Y
o → 14	(14 + 10) mod 26	24 → Y
k → 10	(10 + 10) mod 26	20 → U

Q2

The following shows the character-by-character decryption of the ciphertext. The plaintext is "**book**".

Ciphertext	Decryption	Plaintext
L → 11	(11 – 10) mod 26	01 → b
Y → 24	(24 – 10) mod 26	14 → o
Y → 24	(24 – 10) mod 26	14 → o
U → 20	(20 – 10) mod 26	10 → k

Q3

- Encrypt the message 'this is an exercise' using additive cipher with key = 20. Ignore the space between words. Decrypt the message to get the original plaintext.
- The ciphertext is **NBCMCMUHYRYLWCMY**

Q4

Atbash was a popular cipher among Biblical writers. In *Atbash*, 'A' is encrypted as 'Z', 'B' is encrypted as 'Y', and so on. Similarly, 'Z' is encrypted as 'A', 'Y' is encrypted as 'B', and so on. Suppose that the alphabet is divided into halves and the letters in the first half are encrypted as the letters in the second and *vice versa*. Find the type of cipher and key. Encipher the plaintext 'an exercise' using the *Atbash* cipher.

Answer

- The relationship between the plaintext and ciphertext is P + C = 25 mod 26. In other words, it is monoalphabetic cipher. However, the key is not fixed, it is a stream of odd integers as shown below (P is plaintext, K is key, and C is Cipher text):

P	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
K	25	23	21	19	17	15	13	11	09	07	05	03	01	25	23	21	19	17	15	13	11	09	07	05	03	01
C	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

- Using the above relationship, we can encipher the plaintext as shown below:

Plaintext: anexercise	Ciphertext: ZMVCVIXRHV

Q5

- Alice can use only the additive cipher on her computer to send a message to a friend. She thinks that the message is more secure if she encrypts the message two times, each time with a different key. Is she right? Defend your answer.

Answer

Double encryption here does not help. Encryption with $k1$ followed by encryption with $k2$ is the same as encryption with $k = (k1 + k2) \bmod 26$. Alice has only changed the key. The following shows the proof. P is the plaintext and C is the ciphertext.

C = [(P + k1) mod 26 + k2] mod 26

C = (P mod 26 + *k*1 mod 26 + *k*2) mod 26

C = (P mod 26) mod 26 + (*k*1 mod 26) mod 26 + *k*2 mod 26

C = P mod 26 + *k*1 mod 26 + *k*2 mod 26 = (P + *k*1 + *k*2) mod 26

C = (P + k) mod 26 where *k* = (*k*1 + *k*2) mod 26

Q6

- One of the attacks an intruder can apply to a simple cipher like an additive cipher is called the *ciphertext* attack. In this type of attack, the intruder intercepts the cipher and tries to find the key and eventually the plaintext. One of the methods used in a ciphertext attack is called the *brute-force* approach, in which the intruder tries several keys and decrypts the message until the message makes sense. Assume the intruder has intercepted the ciphertext 'UVACLYZLJBYL'. Try to decrypt the message by using keys from 1 until a plaintext appears that makes sense.

Answer

- We apply the key = 1, 2, 3, 4, 5, 6, and 7 to find the plaintext.

Ciphertext	Key	Plaintext
UVACLYZLJBL	1	tuzbkxeykiaxy
UVACLYZLJBL	2	styajwdxjhzwj
UVACLYZLJBL	3	rsxzivcwigyvi
UVACLYZLJBL	4	qrwyhubvhfxuh
UVACLYZLJBL	5	pqvxgtaugewtg
UVACLYZLJBL	6	opuwfsztfdvsf
UVACLYZLJBL	7	notverysecure

Q7

(i) What is the maximum number of keys that an eavesdropper needs to try in order to unscramble a ciphertext which has been encrypted using the Caesar Cipher?

(ii) What would your answer be if a randomly-ordered cipher alphabet was employed?

(iii) How could we try and crack a cipher based on a randomly-ordered cipher alphabet?

Answer

a) All possible keys of a Caesar cipher can be obtained by shifting the plain alphabet 26 times.

b) If a substitution cipher is used, an eavesdropper has to try up to 26! i.e. 26 x 25 x 24 x ... x 2 x 1 = 26! = 403,291,461,126,605,635,584,000,000 combinations!

c) Frequency analysis can be used to identify the most popular letters in the ciphertext and map them to equally popular letters of the plain alphabet. Grammar and syntax rules of the adopted language can then be used to unscramble the remaining ciphertext.

Q8

A small private club has only 100 members:

a) How many secret keys are needed if all members of the club need to send secret messages to each other?

b) How many secret keys are needed if everyone trusts the president of the club?

c) If a member needs to send a message to another member, she first sends it to the president; the president then sends the message to the other member.

Homework

Q9 - Homework

- Symmetric-key cryptography is analogous to the exchange of a briefcase that contains a gift. The sender uses a padlock to lock the briefcase and hires a courier to deliver it to the intended receiver. A copy of the key is also sent to the receiver via a secure channel. Consider the case of asymmetric-key cryptography, where the sender owns a (red) padlock and a key, whilst the receiver owns a different (blue) padlock and a key. How should the sender have the briefcase delivered to the receiver without the requirement of key exchange?

- Thank you
- Class test next Wednesday on 27/4/2022 @ 7pm